Operando Spectroscopy of Pd and PdAu Laterally Condensed Catalysts during Selective Acetylene Hydrogenation Reaction

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Pd is a highly active catalyst for selective acetylene hydrogenation which attracts industrial and academic interest, since it has high selectivity for ethylene production.(*1*) The selectivity of ethylene can be further improved by incorporation of a second element, e.g. Au as an alloy or C in the near-surface.(*2, 3*) In this study, the surface chemistry and the electronic structure of Pd and PdAu laterally condensed

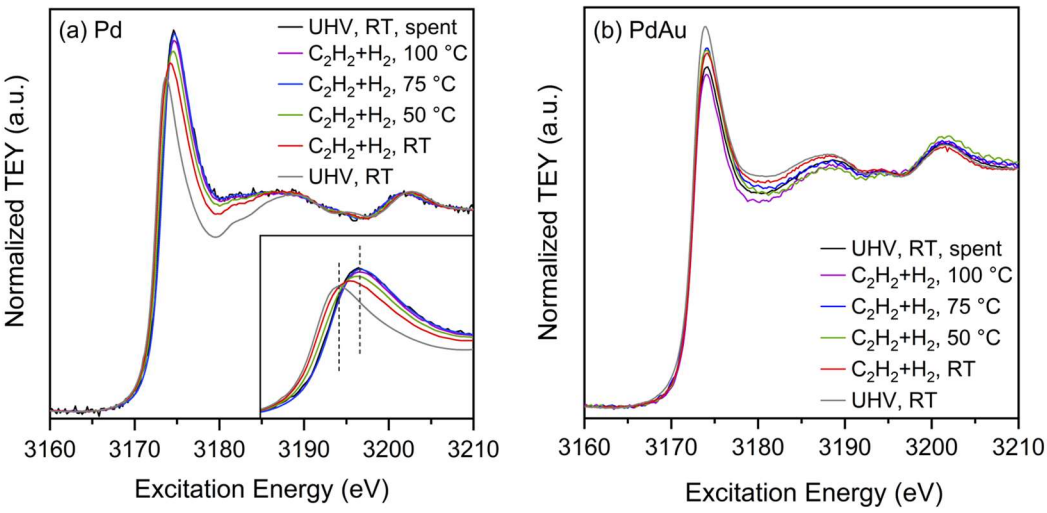

Figure 1. NEXAFS TEY Pd L_3-edge spectra of (a) Pd and (b) PdAu LCC, measured in 1 mbar C_2H_2 and H_2 mixture.

catalysts (LCC) are investigated by means of near ambient pressure X-ray photoelectron spectroscopy (NAP-XPS) and Near Edge X-ray Absorption Fine Structure (NEXAFS) during the selective acetylene hydrogenation reaction. XPS Pd 3d spectra of 3 nm Pd and PdAu LCC samples revealed that the presence of Au inhibits the incorporation of carbon in the Pd lattice, as evidenced by the relative peak intensities of Pd:C (carbon diffused in Pd lattice) and metallic (or alloy) Pd peaks. When the reaction feed (C_2H_2+H_2) is introduced to the NAP-XPS chamber at room temperature, the Pd:C contribution increases in Pd LCC with further increase in reaction temperature. The change in Pd:C contribution is much less in PdAu LCC compared to that of pure Pd LCC throughout the reaction. The inhibition of carbon incorporation into the Pd lattice with the addition of Au in the structure is also validated by Density Functional Theory calculations.

Figure 1 shows NEXAFS Pd L_3-edge spectra of Pd and PdAu LCC during acetylene hydrogenation. In the as-prepared form, both Pd and PdAu has whiteline energy and absorption features that match with that of metallic Pd.(*4*) As the reaction feed is introduced at room temperature, the whiteline shifts to higher energies and broadens in Pd, which indicates formation of the Pd:C phase. When the reaction temperature is increased, the whiteline energy shifts even more and remains constant at and above 75 °C, which agrees with the NAP-XPS experiments. The features related to the Pd:C phase formation is not observed in the Pd L_3-edge spectra of PdAu LCC, validating that Au hinders the carbon incorporation into the Pd lattice.

References

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